

An Unsupervised Learning Approach for Pavement Distress Diagnosis via Siamese Networks

Ruiqi Ren¹, Peixin Shi¹, Pengjiao Jia, and Jinwoo Kim

Abstract—Accurate, automated diagnosis of pavement distress is essential for effective roadway maintenance but presents considerable challenges. Supervised learning methods are constrained by limited labeled data, while existing unsupervised representation learning approaches are difficult to capture the fine-grained details needed for precise pixel-level segmentation in pavement images with similar backgrounds. To address these limitations, we propose a novel unsupervised approach for pavement distress segmentation that employs a new pre-text task within Siamese networks. Our method integrates an explicit prediction head and a high-dimensional cross-entropy loss, enabling implicit class labeling and enhancing fine-grained recognition of distress patterns. Additionally, vision transformers are employed to leverage self-attention mechanisms, facilitating accurate segmentation of foreground distress regions. Experimental results demonstrate that our approach outperforms existing unsupervised representation learning and anomaly detection methods. Notably, when used to pre-train backbone networks such as ResNet-50, our method yields higher accuracy and faster convergence on downstream supervised tasks compared to pre-training on the labeled ImageNet dataset. The proposed method holds promise for advancing pavement maintenance decision-making and enhancing the performance of traditional supervised deep learning models.

Index Terms—Pavement distress, unsupervised deep learning, Siamese networks, vision transformers.

I. INTRODUCTION

ACCURATE automatic pavement distress diagnosis is crucial for efficient roadway maintenance while faces significant challenges [1]. Manual pavement distress diagnosis and maintenance often disrupt road traffic and cost significant labors and time [2]. As a corollary, pavement management must navigate a fine balance between maintenance costs, potential serviceability losses, and structural health preservation [3]. In recent decades, the emergence of fast, economical, and automated pavement distress detection technology has provided an effective solution for pavement management. Integrating a pavement distress diagnosis system onto an inspection vehicle enables automatic diagnoses without disrupting traffic and saves cost [4].

Received 19 December 2023; revised 7 July 2024 and 30 October 2024; accepted 7 November 2024. Date of publication 27 November 2024; date of current version 4 February 2025. This work was supported by the National Natural Science Foundation of China under Grant 52278405. The Associate Editor for this article was Y. Hou. (Corresponding author: Peixin Shi.)

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Digital Object Identifier 10.1109/TITS.2024.3500030

A commonly used pavement distress diagnosis system typically includes a remote sensing module for digital image acquisition, a global positioning system (GPS), and a digital image analysis system for recognition and segmentation. The early image analysis system heavily relies on hand-crafted features, including texture, gradient, and shape descriptors, to identify various distress types [5], [6], [7]. These traditional feature extraction methods have limited generalization capabilities, struggling to accommodate the diverse and complex nature of pavement images. Moreover, designing hand-crafted features is time-consuming, requiring meticulous engineering efforts. There is a pressing need for advanced pavement distress diagnosis techniques that can overcome these limitations and offer more robust, efficient solutions.

In recent years, deep learning has emerged as a promising solution for pavement distress diagnosis [8], [9], [10], [11], [12]. Compared to traditional methods, deep learning is data-driven and can learn to automatically extract relevant features. It eliminates the requirement for laborious manual feature engineering, making it more generalized and suitable for a wide range of diagnosis tasks. Deep learning-based methods represented by supervised learning train a network to recognize pavement distress by feeding it images with manual labels. By sampling different distresses, a deep learning network can be trained to identify cracks, potholes, rutting, and other pavement distresses with high accuracy. However, one of the major challenges of supervised learning is the availability of labeled data. Insufficient data may lead to network overfitting while manual labeling is very labor-intensive.

Deep representation learning offers a novel technique to mitigate the reliance on manual annotations. Mainstream approaches can be broadly categorized into generative learning and contrastive learning. Generative learning leverages the redundant characteristics of image information, allowing for the generation of complete images from a subset of this data, thereby facilitating the extraction of deep representations inherent within the images [13], [14], [15], [16]. Contrastive learning focuses on maximizing the mutual information between different views of the same image, commonly employing Siamese networks [17], [18], [19]. Theoretically, the ability to learn deep representations from pavement images holds the promise of extracting robust distress features without the necessity for manual annotations. However, practical applications reveal several significant challenges. First, distress objects often exhibit unique characteristics, frequently presenting elongated shapes. Consequently, when employing

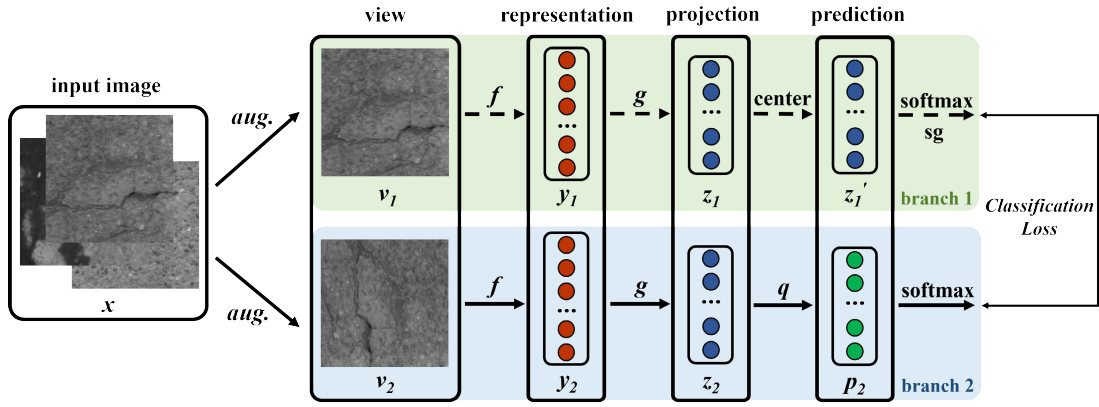


Fig. 1. Training schema of unsupervised siamese networks for pavement distress diagnosis (SiapaveNet).

generative learning architectures with limited information, there is a substantial risk of overlooking critical foreground distress targets, which undermines the learning of robust features; Second, the existing pretext tasks designed for Siamese networks, which are based on the similarity or difference of images within batches, encounter difficulties when applied to pavement images that possess similar backgrounds and small objects; Last, achieving unsupervised distress segmentation through deep representation learning remains a complex challenge that necessitates further exploration. Effectively addressing these limitations is crucial for unlocking the full potential of unsupervised methods in pavement distress segmentation applications.

To address this issue, this study investigates the feasibility of applying unsupervised learning to learn representations directly from unlabeled pavement images. We propose a novel solution for pavement distress diagnosis, which utilizes a predicting classification-based Siamese network with cross-entropy loss and high-dimensional output to provide implicit class labels (SiapaveNet), as illustrated in Figure 1. It utilizes the self-attention mechanism of Vision Transformers (ViTs) to achieve pixel-level segmentation. The proposed framework can be employed to train commonly used backbone networks such as ResNet-50 [20], and the performance is superior to ImageNet pre-trained backbone networks. It is an innovative exploration of fully unsupervised pixel-level pavement distress diagnosis and may serve as a baseline for future research in this field.

II. RELATED WORK

Currently, deep learning is probably the most prevalent and potent technique for pavement distress diagnosis. This section provides an overview of existing deep learning-based methods that may be used for pavement distress diagnosis to highlight the significance and potential of the proposed method.

A. Crack Segmentation

Cracking is the most common form of pavement distress and crack segmentation has received intensive attention in academia. Most existing approaches employ CNNs such as U-Net [21] or Fully Convolutional Networks (FCNs) [22] and

their variants for crack segmentation. For example, Yang et al. [23] proposed a U-Net variant with a pyramid and hierarchical boosting network for pavement crack segmentation. Xiang et al. [24] combined super-resolution image reconstruction with U-Net segmentation to enhance the accuracy of low-pixel microcracks, while Zhang et al. [25] designed a non-symmetric U-Net architecture with a discriminator to effectively segment small cracks. Recently, transformer-based methods have emerged as the state-of-the-art for fine crack segmentation, highlighting the potential of transformers in this field [26]. While current research efforts are mainly focused on improving the accuracy of pavement distress segmentation through supervised learning, there has been very limited work on completely unsupervised approaches. This research aims to explore unsupervised pavement distress diagnosis as an alternative, which has the potential to complement and enhance existing supervised learning solutions.

B. Anomaly Detection

Anomaly detection techniques, commonly utilized for identifying human diseases and industrial defects, can also be adapted to detect pavement distress, which serves as an anomaly on healthy surfaces. While these methods often employ unsupervised or semi-supervised approaches due to the unpredictable nature of anomalies [27], [28], [29], they still rely on weak labels, limiting their effectiveness. Solutions typically involve transforming abnormal images into normal ones for comparative analysis [30] and using various unsupervised techniques such as convolutional autoencoders [31], vector quantized variational autoencoders (VQ-VAE), autoregressive transformers [32], and FastFlow [33]. Recent advancements, such as the Learning Disentangled Priors (LDP) [34], which combines model-driven low-rank representation with data-driven deep learning, have improved anomaly detection capabilities by effectively modeling both background and anomalies through explicit and implicit priors. However, the non-standardized nature of pavement distress images and the influence of diverse field conditions present significant challenges that necessitate the exploration of fully unsupervised feature extraction methods without reliance on weak labels.

C. Unsupervised Representation Learning

Unsupervised representation learning enables deep neural networks to be trained on large unlabeled visual datasets, enhancing their ability to transfer knowledge to specific tasks. One approach involves generative learning methods, which extract information from images by masking most regions and using the visible parts to reconstruct the hidden areas. For instance, Zhou et al. effectively applied Masked Autoencoders (MAE) for medical imaging tasks [15], while Sun et al. introduced RingMo [35], which trains the remote sensing foundation model using masked image modeling. Hong et al. introduced SpectralGPT [16], a 3D generative pretrained transformer specifically designed for spectral remote sensing images, optimizing the analysis of complex datasets. While this technique has been adapted for fields like remote sensing and medical imaging, it faces challenges in pavement distress images, where small targets may be obscured by similar backgrounds, leading to inadequate reconstruction. Another common approach is the use of Siamese networks, which aim to place similar inputs close together in a shared embedding space [36], [37], [38]. Dufumier et al. advanced this approach by incorporating proxy metadata into their contrastive learning framework [39], enhancing positive sampling and improving classification tasks in 3D brain MRI scans. Similarly, Wang et al. proposed TransPath [19], a hybrid model that utilizes self-supervised learning on unlabeled histopathological images, achieving superior performance in downstream classification tasks. However, there is no clear agreement on the best implementation practices, with some methods focusing on predicting outcomes and others on comparing similarities and differences between inputs. In data-limited areas such as pavement distress diagnosis, it is crucial to thoroughly evaluate and choose the most effective unsupervised methods. Integrating deep learning visual explanation methods such as Grad-CAM [40], Ablation-CAM [41], and Score-CAM [42] presents opportunities for enabling pixel-level distress diagnosis via unsupervised learning. However, challenges remain, including whether such methods can achieve fine-grained segmentation for uniquely shaped distresses and whether comparatively small pavement datasets can sufficiently leverage the potential of unsupervised learning. This study investigates these challenges and explores potential solutions.

III. PROPOSED APPROACH

This section describes how the proposed prediction-based Siamese network works and how does it achieve unsupervised pixel-level pavement distress diagnosis.

A. Forward Propagation and Loss Functions

The network architecture, as shown in Figure 1, first extracts an image x from the dataset X , and obtains two different views v and v' through different data augmentation methods t and t' , respectively. Then, v and v' are processed similarly. They are first represented by the backbone network (e.g., ResNet or ViT), and then projected by MLP layers. The difference is that branch 1 of the Siamese network will not update the parameters through back propagation, but through

the momentum encoder. Then branch 2 network will predict the output of branch 1 network by a predictor. The loss function between the resulting vector $z_1 = g(f(v_1))$ and $p_2 = q(g(f(v_2)))$ is cross-entropy:

$$\mathcal{L}(z_1, p_2) = -p_2 \log z_1 \quad (1)$$

After each input of a pair of images, the network updates constantly. To ensure the efficacy of the proposed approach, the Siamese network is treated as a mutual binary classification problem, i.e., to determine if the image input by the Siamese network is augmented by the same image. Therefore, the output vector is softmax processed, modifying Equation (1) as follows:

$$\mathcal{L}(\text{softmax}(z_1), \text{softmax}(p_2)) \quad (2)$$

Since branch 1 no longer uses backpropagation to update its parameters, its gradient is no longer calculated. Therefore, Equation (1) can be expressed as:

$$\mathcal{L}(z_1, \text{stopgrad}(p_2)) \quad (3)$$

Furthermore, a centering operation for the parameters of branch 1 is applied, which is a bias term c added to the network: $g_1(f(v)) \leftarrow g_1(f(v)) + c$. The center c is updated with an exponential moving average:

$$c \leftarrow mc + (1 - m) \frac{1}{B} \sum_{i=1}^B N_1(x_i) \quad (4)$$

where B is the batch size, and N_1 is the Siamese network 1, m is a rate parameter. The pseudo-code implementation of SiapaveNet is in Algorithm 1.

B. Avoiding Collapse

1) *Momentum Encoder*: SiapaveNet incorporates the use of momentum encoders in branch 1 of the networks. The momentum encoder, introduced by [36], is a type of moving average encoder that smoothens the data representations in contrastive learning algorithms by employing exponential moving average over multiple mini-batches of the data representations during the training phase, instead of updating each representation after every mini-batch. This mechanism reduces variance and prevents overfitting. The utilization of momentum encoders in SiapaveNet ensures stability and contributes to the final output result. SiapaveNet employs three networks, f , g , and q , which all make use of the momentum encoder, as summarized by the following equation:

$$\theta_t = \alpha\theta_{t-1} + (1 - \alpha)\theta_t \quad (5)$$

where θ is the network parameter, t represents the iteration step, and α is the smoothing parameter that controls the moving average of the momentum encoder.

2) *Stop-Gradient*: The stop-gradient technique, commonly utilized in conjunction with the momentum encoder, helps in optimizing network by blocking gradients from flowing through certain parts of the network during backpropagation. This helps prevent overfitting and stabilize the training process by preventing updates to positive example representations and

Algorithm 1 SiapaveNet Pseudo-Code, PyTorch-Like

```

# f: backbone
# g: projection mlp
# q: prediction mlp
# tpp, tpz: temperature parameters
# C: center

# load a minibatch x with n samples

for x in SiapaveNet:
    v1, v2 = aug(x), aug(x) # random augmentation
    y1, y2 = f(v1), f(v2) # representations
    z1, z2 = g(y1), g(y2) # projections
    p2 = q(z2) # predictions

    L(p2, z1).backward() # loss and back-propagate
    update(f, h) # SGD update

def L(p, z):
    z = z.detach() # stop-gradient

    p = softmax(p / tpp, dim=1)
    z = softmax((z - C) / tpz, dim=1) # centering

    # cross entropy
    return - (p * log(z)).sum(dim=1).mean()

```

smoothing data representations through exponential moving average, respectively. It is first introduced in by He et al. [36], and the irreplaceable role of stop-gradient has been widely recognized, as demonstrated by Chen and He [38]. SiapaveNet also employs this optimization strategy for ensuring consistency across mini-batches and mitigating overfitting.

3) *Centering*: Inspired by SwAV [43], SiapaveNet employs clustering centers to smooth the output. The clustering centers serve as prototypes to represent clusters of similar data examples. This study calculates the average of the output of each batch of branch 1 as the center. This center affects the output of branch 1 as a bias term. It can be regarded as self-adaption normalization process. The specific operation of centering has been explained in Section III-A and Equation (5).

4) *Batch Normalization*: Batch normalization is widely acknowledged as a pivotal aspect to avoid collapse, as discussed by [37]. It is a normalization technique applied to the activations of neural networks, which helps mitigate the internal covariate shift, i.e., the changes in activation distribution caused by weight updates during the training phase. This technique stabilizes the training process and reduces the sensitivity of the network to the initial weight values. However, it can lead to a loss of discriminative information between image pairs. This study shows its impact in Section V.

C. Unsupervised Distress Segmentation

To achieve unsupervised segmentation of pavement distress, the proposed method utilizes a ViT backbone network (Figure 1), which is a transformer based architecture for image classification tasks, proposed in the seminal work of

Dosovitskiy et al. [44]. The architecture is designed to handle large-scale image data by transforming the input image into a sequence of fixed-length vectors, which are then processed by a Transformer network. Self-attention block is a key component of the Transformer architecture, which enables ViT to capture global information from the image input. Self-attention is implemented by computing attention scores between all vector tokens in the sequence, which can be represented mathematically as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

where Q , K , and V are the query, key, and value matrices, respectively, and d_k is the dimension of the key vectors. The attention scores are then used to weigh the value vectors and generate the output. This mechanism allows the network to attend to different regions of the image and effectively capture long-range dependencies. By combining the self-attention mechanism with the transformer architecture, ViT has achieved state-of-the-art results on several benchmark datasets, demonstrating its effectiveness in large-scale image classification tasks. In the proposed method, the attention mechanism of ViT is aimed to utilize to segment pavement distress without manual labeling. Specifically, the ViT is trained to attend to the foreground of the input image, i.e., pavement distress, by calculating the attention score of each patch of the input image. Later experimental chapters will demonstrate the effectiveness of this approach.

D. Implement Details

1) *Evaluation*: The evaluation metrics utilized by SiapaveNet include mean Dice (mDice), mean Intersection over Union (mIoU), mean Pixel Accuracy (mPA). These metrics are commonly used to evaluate the performance of image segmentation models, providing a comprehensive assessment of the model accuracy and precision in classifying image pixels. The mDice is defined as the average Dice coefficient over all classes, mIoU is the average Intersection over Union over all classes, and mPA is the average Pixel Accuracy over all classes, calculated as:

$$mDice = \frac{2 \sum_{i=1}^C |GT_i \cap Pred_i|}{(|GT_i| + |Pred_i|)/C} \quad (7)$$

$$mIoU = \frac{\sum_{i=1}^C |GT_i \cap Pred_i|}{|GT_i \cup Pred_i|/C} \quad (8)$$

$$mPA = \frac{\sum_{i=1}^C |GT_i \cap Pred_i|}{|GT_i|/C} \quad (9)$$

where C is the number of classes, GT_i is the set of ground truth pixels for class i , and $Pred_i$ is the set of predicted pixels for class i . The Dice coefficient measures the overlap between the predicted and ground truth segments, with a value of 1 indicating perfect agreement.

Mean Average Precision (mAP) is used as an evaluation metric to assess the performance of downstream tasks. The mAP metric calculates the average performance of a detection system across multiple object classes by computing precision and recall values for each class and averaging the results.

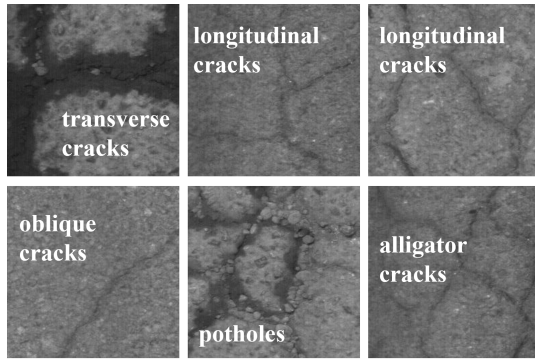


Fig. 2. Examples of pavement distress images following Pre-processing.

2) *Dataset and Other Details*: Two datasets are used to evaluate the pavement distress segmentation capability of the proposed method. One is the continuous pavement images dataset (copave dataset). It comprises of 1,000 large pavement images, acquired via a pavement inspection vehicle. As a pre-processing step, these images are cropped into 88,088 smaller images of size 256×256 pixels each. These images contain various forms of pavement distress, including transverse cracks, longitudinal cracks, oblique cracks, alligator cracks, potholes, and other forms, as shown in Figure 2. Another is the pavementscapes dataset [45], we do the same pre-processing to ensure a comparable results. An A40 GPU is employed for the experiments, with a batch size of 64. AdamW is utilized as the optimization and the learning rate is set to 0.0005. The momentum parameter is set to 0.996.

IV. EXPERIMENT AND RESULTS

In this section, two main outcomes of the proposed approach, known as SiapaveNet, are presented. These include pavement distress segmentation and transfer learning on downstream tasks. To this end, the SiapaveNet framework is firstly employed to train a ViT network, utilizing its self-attention mechanism for unsupervised segmentation. Subsequently, the SiapaveNet framework is utilized to train a ResNet-50 backbone, using it as a pre-trained network for completing downstream tasks. Notably, the training process in the SiapaveNet approach is fully unsupervised.

A. Pavement Distress Segmentation

1) *Baselines*: A comprehensive evaluation of the proposed SiapaveNet method was conducted in comparison to several state-of-the-art unsupervised and semi-supervised techniques with potential applications for pavement distress segmentation. As few pavement distress diagnosis approaches are based on unsupervised or semi-supervised learning, a set of methods demonstrating promising results in fields like human disease detection, industrial defect identification, and other cutting-edge unsupervised representation learning were selected. These included f-Anogan [27], FastFlow [33], GANormaly [29], MAE (Masked Autoencoders) [13] and BEIT (Bidirectional Encoder representations from Image Transformers) [14]. f-AnoGAN, GANormaly, and FastFlow are anomaly detection methods that consider only normal

TABLE I
COMPARISON OF SIAPAVENET WITH EXISTING METHODS
ON COPAVE DATASET

Method	mDice	mIoU	mPA
FastFlow [33]	0.138	0.090	0.784
f-Anogan [27]	0.252	0.165	0.748
GANormaly [29]	0.103	0.056	0.697
MAE [13]	0.186	0.107	0.559
BEIT [14]	0.339	0.218	0.609
SiapaveNet(Ours)	0.523	0.366	0.807

data during training, with an auto-encoder reconstructing images. When abnormal images are input, the auto-encoder cannot reconstruct anomalous regions, enabling their detection. For this experiment, these baselines utilized only normal pavement images during training. Conversely, MAE and BEIT are unsupervised representation learning approaches that learn effective image representations from unlabeled data. MAE randomly masks out patches of the input image and is trained to reconstruct the original unmasked image. BEIT is a Transformer-based approach that learns representations by predicting both the masked image patches and the visual tokens obtained by performing various image transformations.

2) *Results on Copave Dataset*: Table I and Figure 3 demonstrate the superior performance of the proposed method for pavement distress segmentation. It outperforms most unsupervised and semi-supervised methods, while requiring less manual labeling effort. The method achieves an mDice of 0.523, mIoU of 0.366, and mPA of 0.807. Both mDice and mIoU provide a balanced evaluation by considering the overlap between the predicted and ground truth masks, with mDice being more sensitive to instances with small areas, while mIoU can be more influenced by instances with larger areas. Meanwhile, mPA computes the ratio of correctly classified pixels averaged across instances, which may overestimate performance for small targets due to the dominance of background pixels. In Table I, FastFlow obtains the highest mPA, but it only assumes that the image does not contain pavement distress, and its effect is obviously inferior to that of SiapaveNet. Figure 3 provides further visual evidence of the effectiveness of the proposed method in achieving accurate pixel-level pavement distress diagnosis. SiapaveNet can accurately locate the distress pixels and differentiate pavement distress from the background. While other anomaly detection schemes such as f-AnoGAN and GANormaly can obtain the basic contour of pavement distress, they are weaker than SiapaveNet across various metrics. MAE and BEIT are deep representation learning methods using the same backbone network structure and self-attention mechanism as the proposed method for segmentation. BEIT has shown better performance than other anomaly detection schemes but weaker than the proposed SiapaveNet. However, the masking patches and reconstructing of MAE may not be well-suited for small pavement distress instances, resulting in bad performance and weaker than most baselines. These results suggest that the proposed SiapaveNet method is a highly effective approach for pavement distress segmentation and presents a viable alternative for semi-supervised anomaly detection methods.

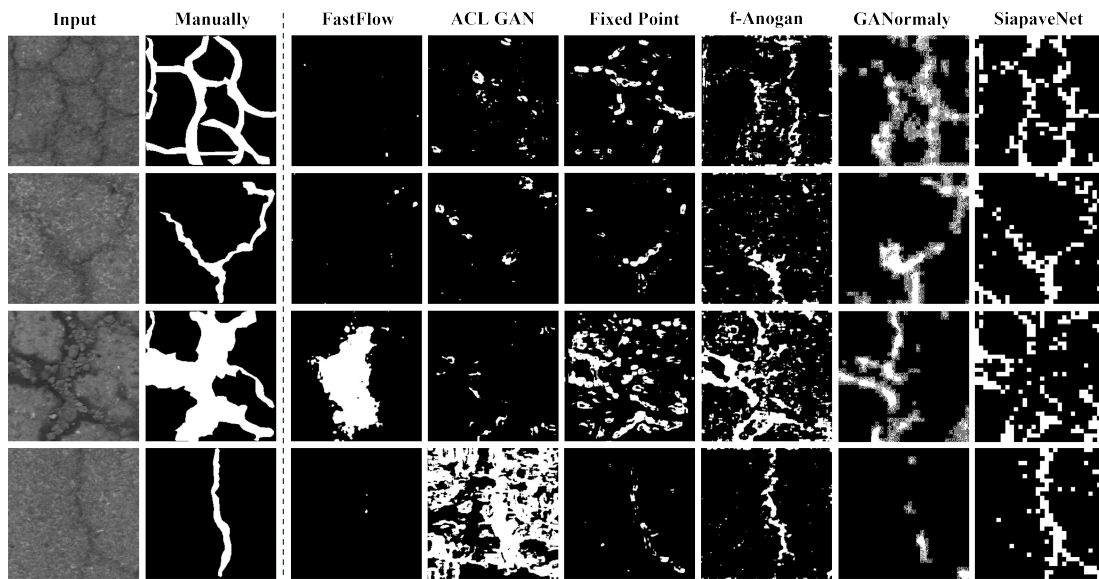


Fig. 3. Comparative analysis of SiapaveNet with existing approaches for pavement distress segmentation.

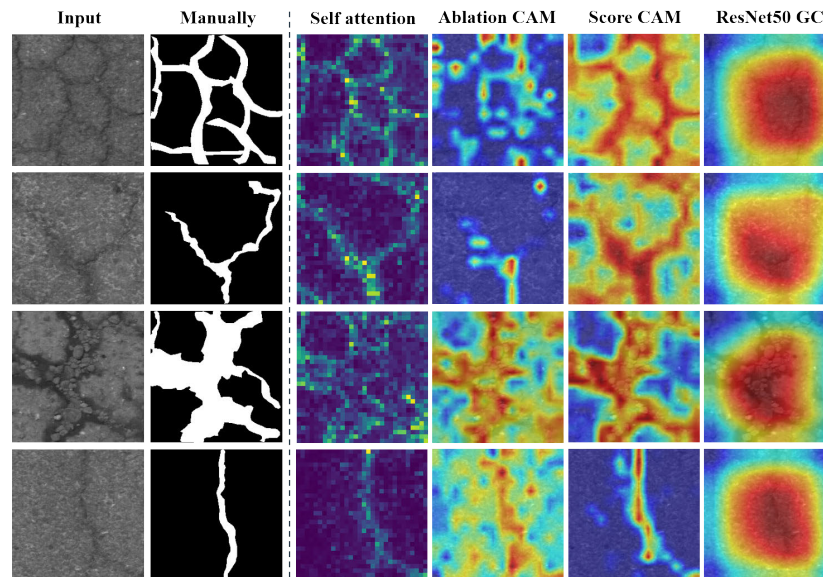


Fig. 4. Comparison of Self-attention methods with visual explanation methods for pavement distress segmentation.

Compared with anomaly detection methods, SiapaveNet is an unsupervised approach, which do not use image-level labels. As a result, it cannot segment pavement distress via converting abnormal images into normal ones. SiapaveNet uses a self-attention visualization scheme. It is compared with other visual explanation techniques in Figure 4, including Grad-CAM and its variants. It is observed that the self-attention visualization scheme more accurately segments pavement distress when ViT is used as the backbone network of SiapaveNet. To investigate the performance of CNNs without self-attention heads, experiments are conducted using ResNet-50 and show regions of interest using Grad-CAM, as illustrated in the last column of Figure 4. The results indicate that CNNs can only focus on more obscure ranges, highlighting the advantages of self-attention and ViT.

Figure 5 displays the specific attention of different self-attention head. The brightness of the color serves as the score of the attention head for that patch. The observations reveal that not all self-attention heads can accurately focus on the foreground object (i.e., the pavement distress). Specifically, self-attention head 0 and 3 show more attention to the foreground object, while self-attention head 4 shows more attention to the background (i.e., the pavement). Self-attention head 1, 2 and 3 are ambiguous. This pattern is observed in all test images, leading to use self-attention head 3 as the primary tool for segmenting pavement distress. This finding specifically addresses the second challenge raised earlier, which is how to extract pavement distress characteristics and distinguish foreground objects from the background.

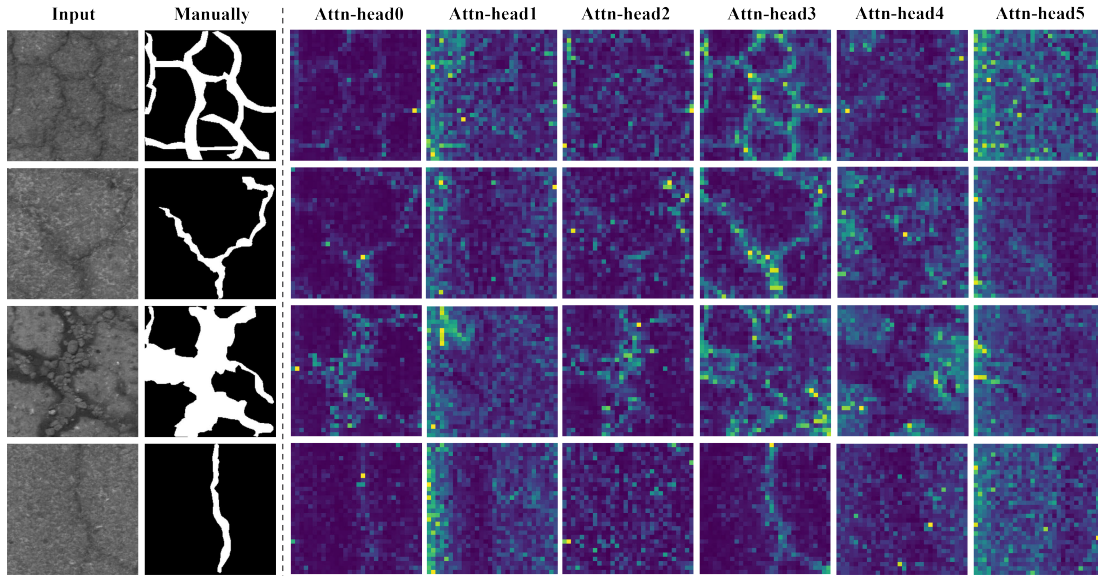


Fig. 5. Visualization of Self-attention mechanism in pavement distress segmentation.

TABLE II
COMPARISON OF SIAPAVENET WITH EXISTING METHODS ON
PAVEMENTSCAPES DATASET

Method	mDice	mIoU	mPA
FastFlow [33]	0.059	0.035	0.831
f-Anogan [27]	0.206	0.128	0.536
GANormaly [29]	0.133	0.089	0.760
MAE [13]	0.284	0.193	0.669
BEIT [14]	0.348	0.244	0.753
SiapaveNet(Ours)	0.439	0.309	0.803

3) *Results on Pavementscapes Dataset:* We also evaluated SiapaveNet against other methods using the pavementscapes dataset. The results are presented in Table II. Our proposed method outperforms other approaches across two evaluation metrics, demonstrating its advanced capabilities. Notably, traditional anomaly detection methods require manual classification of normal data, as the current dataset contains few images of normal pavement. This process typically involves manually labeling and training a deep learning classifier, which may not perfectly capture the normal class. While we strive to exclude distress images from the training data, manual screening of large datasets is impractical, highlighting the advantage of our proposed fully unsupervised learning scheme. The results in Table II align with those in Table I, underscoring the consistency of the superior performance across different datasets. Existing anomaly detection methods continue to perform poorly on this dataset, while BEIT achieves the second-highest segmentation accuracy after our proposed SiapaveNet.

B. Transfer Learning on Downstream Tasks

1) *Tasks and Datasets:* This section presents the outcomes of finetuning a pre-trained ResNet-50, acquired from the unsupervised SiapaveNet framework, on three distinct downstream

tasks related to pavement distress diagnosis, namely pavement crack semantic segmentation, object detection, and instance segmentation. The pavement crack semantic segmentation task entails segmenting cracks from pavement, while the object detection task focuses on locating cracks on pavement. The instance segmentation task aims to locate and roughly segment the cracks on pavement.

These tasks are illustrated in Figure 6. To evaluate the performance of the proposed method, three different datasets are used for each task: dataset I for semantic segmentation [23], dataset II for object detection [46], and dataset III for instance segmentation [47]. These datasets consist of 1896, 1538, and 133 images, respectively with dimensions of 640×360 , 1612×1947 , and 480×320 . They are different from the dataset used in pre-training.

2) *Results:* The evaluations presented in Figures 7, 8, and 9 demonstrate the effectiveness of the proposed approach in the downstream tasks of semantic segmentation, object detection, and instance segmentation. The metrics used for semantic segmentation are mIoU, while mAP is used for both object detection and instance segmentation. The results indicate that the proposed method outperforms the training-from-scratch method in terms of both convergence speed and accuracy in all three tasks. When compared to networks pre-trained on labels and a larger dataset such as ImageNet, the proposed approach demonstrates a faster convergence speed and comparable accuracy in semantic segmentation and instance segmentation, while slightly lower performance in object detection. This discrepancy in performance may be attributed to the larger image size and increased information content of the dataset used for object detection.

V. ABLATION STUDY

This section presents the results of comprehensive ablation experiments that investigate the impact of different settings on the performance of SiapaveNet. The aim is to analyze the

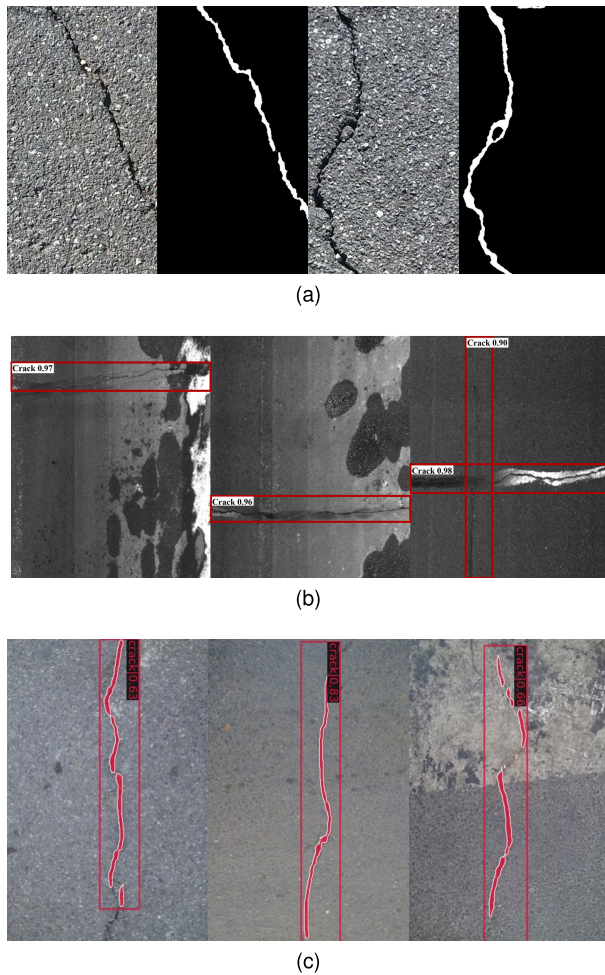


Fig. 6. Exploring three downstream tasks for pavement distress diagnosis: (a) Semantic segmentation, (b) Object detection, and (c) Instance segmentation.

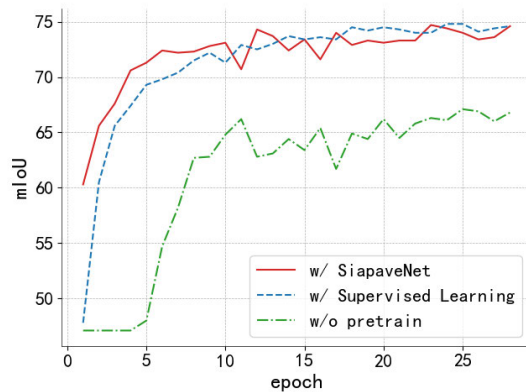


Fig. 7. Semantic segmentation evaluation.

significance of each component of the system and how they affect its overall performance.

A. Comparison With Similar Methods

Our primary objective is to investigate the potential of unsupervised representation learning for pavement distress segmentation. Previously, we demonstrated that our proposed scheme outperforms generative learning models such as MAE

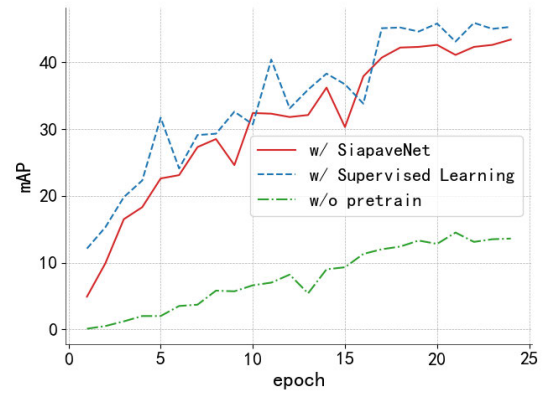


Fig. 8. Object detection evaluation.

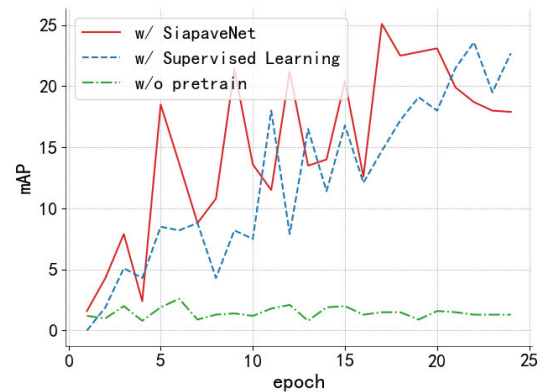


Fig. 9. Instance segmentation evaluation.

and BEiT. In this section, we compare various Siamese network methods that are conceptually similar to our proposed approach, as illustrated in Figure 10. Specifically, we examine:

(1) MoCov3 [17]. It employs a prediction head and the InfoNCE loss to compare the outputs of the Siamese network branches;

(2) BYOL [37]. It utilizes a prediction head and the mean squared error (MSE) loss to regress the output of one Siamese network branch to the other;

(3) SimSiam [38]. It uses a prediction head and the negative cosine similarity (NCS) loss, without a momentum encoder, to measure the similarity between the output vectors of the Siamese network branches;

(4) DINO [18]. It directly predicts the output of one Siamese network branch from the other under cross entropy (CE) loss without employing a prediction head;

(5) SiapaveNet (Ours). It incorporates a prediction head to predict a vector from one branch of the Siamese network based on the other branch under CE loss.

The results demonstrate that employing a prediction head (predictor) combined with a cross-entropy loss function for the classification task achieves the most effective performance as shown in Table III. The proposed SiapaveNet method outperforms similar schemes across all evaluation metrics. Existing self-supervised learning methods face limitations in pavement distress detection tasks due to the subtle differences between

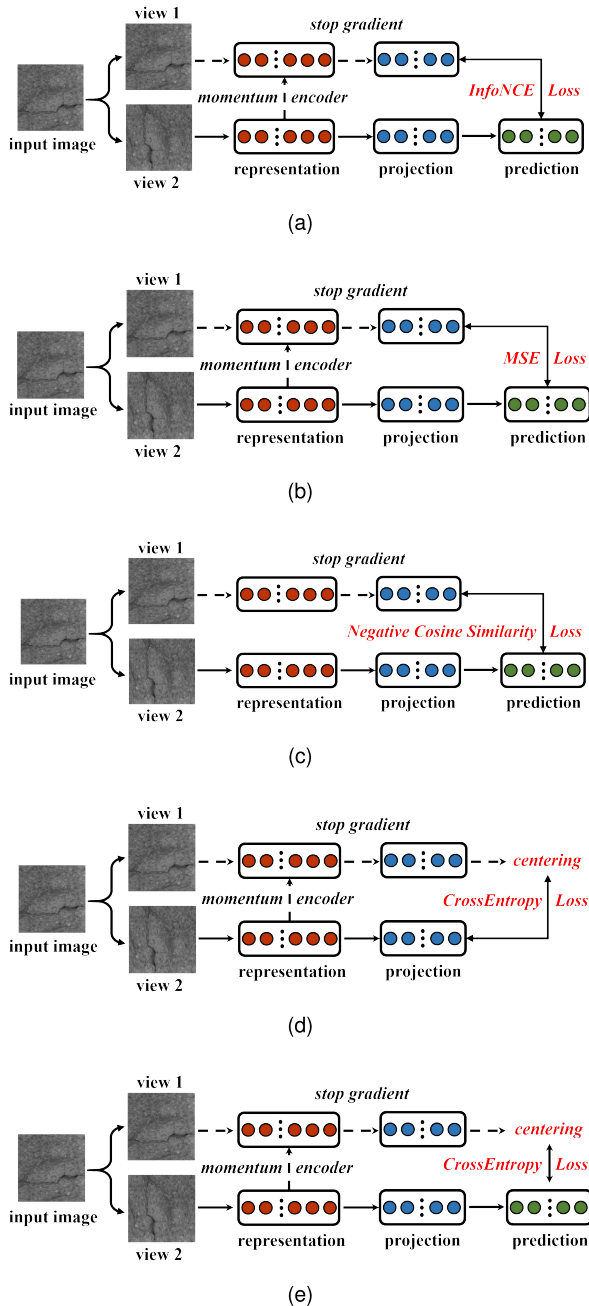


Fig. 10. Training pipeline of different methods: (a) MoCo v3, (b) BYOL, (c) SimSiam, (d) DINO, and (e) SiapaveNet(ours).

foreground objects and consistent backgrounds. MoCo v3 employs the InfoNCE loss, a contrastive learning function that maximizes mutual information between different views of the same image while minimizing it for different images. However, this approach struggles to capture fine-grained distinctions when images within a batch are highly similar. BYOL uses a regression-based loss, predicting the consistency of the output from other branch, which may lack the detail needed for distinguishing subtle distress. The NCS loss in SimSiam, focusing on feature vector similarity, potentially provides insufficient supervision for subtle foreground differences. DINO utilizes high-dimensional outputs and implicit labels via softmax for cross-entropy loss, allowing direct comparison

of probability distributions. While this enables complex relationship capture, its symmetric branch structure may limit fine-grained feature separation. Our proposed SiapaveNet addresses these challenges by employing an explicit prediction head architecture with cross-entropy loss and high-dimensional output. This approach provides implicit class labels and detailed gradient information, enabling more detailed representation of subtle distress patterns. The cross-entropy loss encourages the model to distinguish between different distress classes, even when visually similar. The asymmetric Siamese structure, combined with the prediction task, facilitates learning of invariant features while maintaining sensitivity to fine-grained differences.

B. Different Settings on Predictor

1) *Number of MLP Layers*: One crucial hyperparameter of the multi-layer perceptron (MLP) used as the predictor is the number of layers. This section evaluates the influence of MLP depth on the performance of the model while keeping the previously determined optimal settings fixed. The results, presented in Table IV, indicate that the best performance is achieved when employing a 3-layer MLP as the predictor.

2) *Output Dimension*: Another crucial parameter is the output dimension of the MLP prediction head, especially when combined with the cross-entropy loss function. As demonstrated in Table V, a larger output dimension enables the model to learn more detailed information, but at the cost of increased computational overhead.

3) *Batch Normalization*: While previous study [37] suggests that batch normalization is crucial for maintaining model stability and enhancing performance, our findings indicate that excluding batch normalization from the prediction head yields better results for the proposed model, as shown in Table VI.

C. Loss Functions

As mentioned earlier, the results demonstrate that classification tasks employing a combination of a prediction head (predictor) and a cross-entropy loss function yield the most effective performance. However, the results also compare the differences between various networks. In this section, we utilize the exact same network architecture to evaluate the impact of the loss function on the results. Specifically, we compare three commonly used loss functions: InfoNCE loss, negative cosine similarity (NCS) loss, and cross-entropy (CE) loss. Interestingly, when we fixed the optimal settings obtained in the previous section, only the CE loss and NCS loss enabled the model to function properly as shown in Table VII. InfoNCE failed to decrease the loss, resulting in a collapse of the self-supervised mapping. This collapse manifested as each image patch exhibiting the same value, indicating a failure in learning meaningful representations.

D. Centering

Centering is an important technique that ensures stable convergence of the network. For the predicting classification task of the proposed method, it is necessary to maintain sufficient stability in the predicted network. In fact, the predicted

TABLE III
COMPARISON WITH OTHER SIMILAR METHODS

Methods	Pred.	Loss	Center.	Momen.	mDice	mIoU	mPA
Moco v3 [17]	✓	InfoNCE	–	✓	0.337	0.215	0.536
BYOL [37]	✓	MSE	–	✓	0.479	0.335	0.752
Simsiam [38]	✓	NCS	–	–	0.332	0.224	0.705
DINO [18]	–	CE	✓	✓	0.507	0.347	0.769
SiapaveNet(Ours)	✓	CE	✓	✓	0.523	0.366	0.807

TABLE IV
COMPARISON OF DIFFERENT NUMBER OF MLP LAYERS

MLP Layers	mDice	mIoU	mPA
2	0.230	0.155	0.708
3	0.523	0.366	0.807
4	0.484	0.332	0.806

TABLE V
COMPARISON OF DIFFERENT OUTPUT DIMENSION

Output Dim.	mDice	mIoU	mPA
256	0.365	0.234	0.753
1024	0.401	0.289	0.657
4096	0.470	0.327	0.757
16384	0.523	0.366	0.807
65536	0.517	0.356	0.777

TABLE VI
COMPARISON ON BATCH NORMALIZATION

Proj. BN	mDice	mIoU	mPA
✓	0.393	0.260	0.753
–	0.523	0.366	0.807

TABLE VII
COMPARISON OF DIFFERENT LOSS FUNCTIONS

Loss	mDice	mIoU	mPA
NCS	0.457	0.308	0.761
InfoNCE	–	–	–
CE	0.523	0.366	0.807

TABLE VIII
COMPARISON ON CENTERING

centering	mDice	mIoU	mPA
–	0.387	0.225	0.754
✓	0.523	0.366	0.807

network utilizes stop-gradient and centering techniques, and its parameters are updated through the momentum encoder. This configuration significantly enhances performance of the model, as evidenced by the results presented in Table VIII.

E. Optimizer

This section aims to evaluate the performance of the proposed SiapaveNet when trained using stochastic gradient descent (SGD) and adaptive moment estimation with

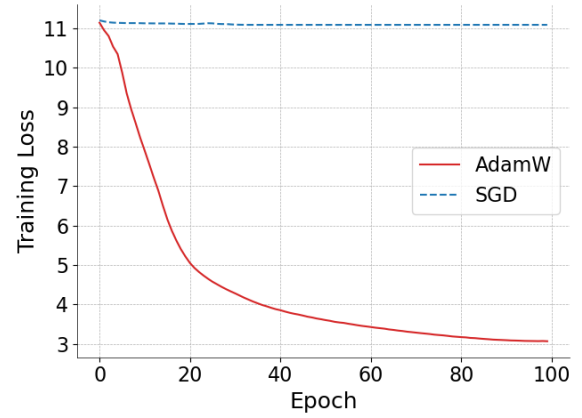


Fig. 11. Comparison of loss convergence between SGD and AdamW optimizers.

decoupled weight decay (AdamW) optimizer. The results show that the model does not converge when trained using SGD, as evidenced by the non-decreasing trend in the loss curve shown in Figure 11. This suggests that the choice of optimization algorithm has a significant impact on the performance of the SiapaveNet. Compared to SGD, AdamW is found to be more effective in optimizing the SiapaveNet, as indicated by the decrease in loss over the course of training. This outcome is consistent with the findings of prior studies [48], [49], which have shown that Adam and its variants such as AdamW and AMSGrad can outperform SGD in the training of visual representations. This difference in performance is attributed to the nature of the ViT network, which has been shown to be less prone to overfitting. The faster convergence of the network with AdamW may therefore be a result of its ability to reduce the loss more effectively in such a scenario. This hypothesis is further supported by the successful convergence of the ResNet-50 when trained with SGD, as presented in Section IV-B.2.

VI. DISCUSSION

Unsupervised deep learning represents a promising direction for the future advancement of pavement distress diagnosis. It has achieved significant success in natural language processing, general computer vision, and other fields due to the scarcity of labeled data compared to the vast amount of unlabeled data available. For the task of pavement distress diagnosis, unsupervised learning is a critical tool to address the challenge of expensive and limited labeling of pavement distress types. This study makes an exploratory effort to assess the potential and limitations of unsupervised pavement distress diagnosis.

An important aspect of this study is the effects of different techniques on the stable convergence and performance of unsupervised pavement distress diagnosis methods. The unique shape of the pavement distress object makes ensuring stable convergence a difficult problem. Technically, we aim to identify specific measures within the Siamese network framework to achieve better pavement distress segmentation performance. Comprehensive ablation experiments demonstrate that employing a predicting classification task using a prediction head and cross-entropy loss function is more suitable for this task compared to other existing methods. We posit that ensuring stable variation in the predicted branch while maintaining flexibility in the predicting branch is a crucial element. This could explain why the predicting branch requires a prediction head, needs to be updated by back-propagation without batch normalization, while the predicted branch should be kept as stable as possible, updated with a momentum encoder, and incorporate stop-gradient and centering techniques. Besides, we speculate that a larger output dimension of the prediction head incentivizes the network to learn more detailed information. We hypothesize that if the classification task merely requires determining whether the predicting branch belongs to the same class as the predicted branch, the network may easily converge to suboptimal solutions or learn shortcut strategies. In addition, it is observed that the AdamW optimizer performs better when training ViT networks, while the SGD optimizer performs better when training CNN networks, similar to supervised learning. These findings provide valuable insights for future unsupervised pavement distress diagnosis methods.

It is a significant advantage that the unsupervised learning-based approach can identify rare types of distress that are difficult to label beforehand, when faced with a small sample of tasks [27], [29], [30]. In addition, unsupervised learning may have a greater advantage when dealing with tasks with limited samples. The experimental results show that when trained with only 133 images over 5 epochs, the accuracy of the unsupervised pre-training model is more than double that of the supervised pre-training model commonly used in the field. Several studies [17], [36] have suggested that the distribution of unsupervised and supervised learning differs significantly, with unsupervised learning having the ability to identify distinct clusters or patterns, which is advantageous when dealing with the unusual form of pavement distress. This may also explain the results of the semantic segmentation task, as shown in Figure 7. When faced with the object detection task in Figure 8, our results are not as good as those obtained using the supervised pre-training model based on ImageNet. The large pixel size of the dataset for this task (nearly 50 times that of the pre-training image), the substantial difference between the object detection and the object segmentation task, and the presence of a specific target, pavement irrigation sewing, in the target dataset may be contributing factors.

The dataset used in this study comprises pavement images captured by continuous sections, totaling 88,000 small images, which represents a significant improvement over previous methods in terms of the number of images that can be processed. However, there is still a considerable gap compared to

datasets in other fields such as the COCO dataset [50], which contains 330,000 images. Our work makes efforts to expand the dataset by using common data augmentation methods. And it is demonstrated that using artificially generated data with weak labels is possible to expand datasets according to recent studies [18], [51], [52]. This may be an effective solution and our future research direction.

Although pixel-level unsupervised deep learning is not yet fully mature for direct application in real-world pavement distress diagnosis projects due to its current limitations, it has shown promising potential for improving downstream tasks. Furthermore, even with limited evaluation, it can provide valuable insights for decision-making in pavement maintenance systems without relying on time-consuming manual labeling. Moving forward, larger and more comprehensive datasets, advanced data augmentation techniques, and more powerful models are expected to significantly enhance the accuracy of pixel-level unsupervised deep learning in pavement distress diagnosis.

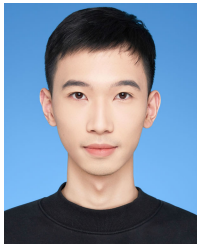
VII. CONCLUSION

This study proposes a novel self-supervised approach for pixel-level pavement distress diagnosis, effectively addressing the challenges of extracting fine-grained features in complex and similar backgrounds. By employing a prediction task with an explicit prediction head and integrating ViTs with self-attention mechanisms, the method enables precise distress segmentation without the need for labeled data. Extensive experimental results demonstrate the superior performance of this approach over existing unsupervised representation learning and anomaly detection baselines. Furthermore, as a pre-training strategy, the proposed method significantly enhances the accuracy and convergence speed of backbone networks in downstream pavement distress diagnosis tasks compared to conventional supervised pre-training on ImageNet. This approach offers a promising pathway for improving pavement maintenance decision-making through reliable unsupervised distress detection. It facilitates more effective identification and prioritization of repair needs, thereby extending infrastructure longevity and reducing maintenance costs.

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